DEMOGRAPHY

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Demography = numbers game! Learn the rates and India data!

Demography | PSM | 3rd Year | MBBS

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The demographic transition theory is asked in every exam!

DEMOGRAPHY

Scientific study of human populations — size, structure, distribution, and changes (births, deaths, migration).

Demographic Transition Theory (4 Stages)

Stage	Birth Rate	Death Rate	Growth Rate	Example
Stage 1	High	High	Low/Stable	Primitive societies
Stage 2	High	Declining	Rapid increase	India (transitioning)
Stage 3	Declining	Low	Slowing growth	Brazil, China
Stage 4	Low	Low	Zero/Negative	Japan, Germany, Sweden

★ India is in LATE Stage 2 / Early Stage 3 — death rate fallen, birth rate declining!

Sources of Demographic Data in India

Source	Conducted By	Key Features
Census	Registrar General of India	Every 10 years; since 1872; de facto method
SRS	Registrar General of India	Sample Registration System — BR, DR estimates
CRS	Local bodies	Civil Registration System — births & deaths
NFHS	MoHFW + IIPS	National Family Health Survey (latest: NFHS-5)
DLHS	MoHFW + IIPS	District Level Household Survey
NSS	NSSO (MoSPI)	National Sample Survey — socioeconomic data

Census of India — Key Facts

- First Census: 1872 (non-synchronous); First regular: 1881 (every 10 years since)
- Method: De facto (count where found) — NOT de jure (permanent residence)
- Latest: Census 2011 — Population: 121.08 crore (1.21 billion); next due 2021 (delayed)
- Census Commissioner: Under Registrar General & Census Commissioner of India

Vital Rates — CBR, CDR, IMR, MMR, TFR

THESE formulas and Indian values come in EVERY. SINGLE. EXAM!

Vital Statistics — Formulas + Indian Data

Rate	Formula	India (SRS 2020)	Significance
CBR	Live births / Mid-yr pop x 1000	19.5 per 1000	Fertility indicator
CDR	Deaths / Mid-yr pop x 1000	6.0 per 1000	Mortality indicator
IMR	Infant deaths (<1yr) / Live births x 1000	28 per 1000	Best health indicator
NMR	Neonatal deaths (<28d) / Live births x 1000	20 per 1000	Neonatal care quality
PMR	Perinatal deaths / Total births x 1000	22 per 1000	Obstetric + neonatal
MMR	Maternal deaths / Live births x 100,000	97 per 100,000	Maternal care quality
TFR	Avg children per woman (lifetime)	2.0	Replacement = 2.1

IMR Components (Most Important Breakdown)

Component	Period	Main Causes
Early Neonatal	0-7 days	Birth asphyxia, prematurity, LBW, sepsis
Late Neonatal	7-28 days	Neonatal sepsis, pneumonia, diarrhoea
Post-neonatal	28 days - 1 yr	Diarrhoea, pneumonia, malnutrition

★ **KEY RELATIONSHIPS:**

Natural Growth Rate = CBR - CDR | Doubling Time = 70 / Growth Rate (%)

Replacement level TFR = 2.1 (India achieved ~2.0 in NFHS-5 — below replacement!)

Other Fertility Measures

Measure	Formula / Definition	Key Point
GFR	Births / Women 15-49 x 1000	More refined than CBR
NRR	Daughters surviving to maternal age	NRR = 1 = exact replacement
GRR	Daughters born per woman (lifetime)	GRR > NRR always
Couple Prot. Rate	% couples using contraception	India: ~67% (NFHS-5)

Pyramid types and India's Census data are viva favourites!

Types of Population Pyramids

Shape	Type	Features	Example
Broad base	Expansive	High BR, high young pop	India, Nigeria, Pakistan
Narrow base	Constrictive	Low BR, aging pop	Japan, Germany, Italy
Even sides	Stationary	BR = DR, stable pop	Sweden, Australia

India's Demographic Profile (Census 2011 / NFHS-5)

Parameter	Value
Population (Census 2011)	121.08 crore (1.21 billion)
Decadal growth (2001-2011)	17.64%
Sex ratio (overall)	943 females per 1000 males
Child sex ratio (0-6 yrs)	914 females per 1000 males
Literacy rate	74.04% (Male 82.1%, Female 65.5%)
Urban population	31.2% (Census 2011)
Population density	382 per sq km
Dependency ratio	Young (0-14): 29%; Old (60+): 8.6%
Life expectancy at birth	Male 69.4, Female 72.0 yrs (SRS 2018-22)
Demographic dividend	Working age (15-64) = ~67% of population

★ DEMOGRAPHIC DIVIDEND (Hot Topic!)

India has world's largest youth population — window of opportunity until 2055!

If invested in education + health + skills = economic boom. If not = burden.

★ **India Sex Ratio: 943 (overall) < 914 (child) = declining!**

Kerala = best sex ratio (1084). Haryana = worst (879). Remember extremes!

Family planning methods and NPP 2000 are asked almost every year!

Family Planning Methods — Classification

Category	Method	Key Details
Spacing	OCP (oral pills)	Combined (estrogen+progestin); Mala-N (free in India)
	IUD / IUCD	Cu-T 380A = 10 yrs; most popular spacing method in India
	Condom (male)	Nirodh — free in India; also STI protection
	Injectable	DMPA (Antara) — 3 monthly injection; newer in India
Terminal	Tubectomy (female)	Most popular method overall in India (36%)
	Vasectomy (male)	Simpler, safer but less accepted in India
Natural	LAM	Lactational amenorrhoea — up to 6 months postpartum
	Rhythm / SDM	Avoid fertile period — Standard Days Method
Emergency	Levonorgestrel	EC pill within 72 hrs; Cu-T within 5 days

National Population Policy 2000

Immediate:	Address unmet need for contraception, maternal & child care
Medium-term:	TFR = 2.1 by 2010 (replacement level)
Long-term:	Stable population by 2045 (at 1.63 billion)
Key strategies:	Promote male participation, delay marriage, spacing methods
Incentives:	Promote 2-child norm; Balika Samridhi Yojana for girl child
Current status:	TFR reached 2.0 (NFHS-5, 2019-21) — below replacement!

India's Family Planning Programme — Timeline

Year	Milestone
1952	India = FIRST country in world to adopt national FP programme
1977	Name changed: Family Planning -> Family Welfare (post-Emergency)
1997	RCH (Reproductive & Child Health) programme launched
2000	National Population Policy 2000
2005	NRHM launched (now NHM) — integrated FP + MCH + disease control
2017	Mission Parivar Vikas — 146 high-fertility districts targeted

Numbers game, Sammy!
Memorise the Indian data
and you'll ace this!

Most Repeated LAQs (10-12 marks)

- 1. Define demographic transition. Describe the 4 stages with Indian context
- 2. Define and explain vital statistics — CBR, CDR, IMR, MMR, TFR with formulas
- 3. Describe family planning methods available in India (classification + details)
- 4. Write about National Population Policy 2000 — objectives and strategies
- 5. Describe components of IMR. Why is it the best health indicator?

Most Repeated SAQs (5-6 marks)

- 1. Sources of vital statistics in India
- 2. Types of population pyramids with examples
- 3. TFR and NRR — define and significance
- 4. Census of India — method and key facts
- 5. Demographic dividend — explain with Indian context
- 6. Emergency contraception — methods and timing

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Best indicator of health status	IMR (Infant Mortality Rate)
India's IMR (SRS 2020)	28 per 1000 live births
India's MMR (SRS 2018-20)	97 per 1,00,000 live births
India's TFR (NFHS-5)	2.0 (below replacement level 2.1)
Replacement level TFR	2.1
Most popular FP method in India	Tubectomy (female sterilisation)
First country with national FP programme	India (1952)
Census method in India	De facto (count where found)
Best sex ratio state	Kerala (1084)
Demographic dividend window — India	Until ~2055
MMR denominator	Per 1,00,000 live births (NOT per 1000!)
NRR = 1 means	Exact replacement (stable population)

♥ Every number tells a story about a life, Sammy!
You're learning to understand India's health story. Keep going!